

Day 14 MYSQL ASSIGNMENT

```
USE slatask;

CREATE TABLE prod14(
product_id INT PRIMARY KEY AUTO_INCREMENT,
name VARCHAR(100),
price DECIMAL(10,2),
stock_quantity INT,
status VARCHAR(20) DEFAULT 'Available'
);

CREATE TABLE audit14(
audit_id INT PRIMARY KEY AUTO_INCREMENT,
product_id INT,
action_type VARCHAR(50),
old_value VARCHAR(100),
new_value VARCHAR(100),
changed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 1 Prevent Negative Stock

DELIMITER $$

CREATE TRIGGER trg_neg_stock
BEFORE UPDATE ON prod14
FOR EACH ROW
BEGIN
IF NEW.stock_quantity < 0 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT='Stock cannot be negative';
END IF;
END$$

DELIMITER ;

-- 2 Audit Price Changes

DELIMITER $$
```

```
CREATE TRIGGER trg_price_update
AFTER UPDATE ON prod14
FOR EACH ROW
BEGIN
IF OLD.price <> NEW.price THEN
INSERT INTO audit14(product_id,action_type,old_value,new_value)
VALUES(OLD.product_id,'Price Update',OLD.price,NEW.price);
END IF;
END$$
```

```
DELIMITER ;
```

```
-- 3 Auto Update Status Out of Stock
```

```
DELIMITER $$
```

```
CREATE TRIGGER trg_out_stock
BEFORE UPDATE ON prod14
FOR EACH ROW
BEGIN
IF NEW.stock_quantity=0 THEN
SET NEW.status='Out of Stock';
END IF;
END$$
```

```
DELIMITER ;
```

```
-- 4 Uppercase Product Names
```

```
DELIMITER $$
```

```
CREATE TRIGGER trg_upper_name
BEFORE INSERT ON prod14
FOR EACH ROW
BEGIN
SET NEW.name=UPPER(NEW.name);
```

END\$\$

DELIMITER ;

-- 5 Log New Product Additions

DELIMITER \$\$

```
CREATE TRIGGER trg_new_product
AFTER INSERT ON prod14
FOR EACH ROW
BEGIN
INSERT INTO audit14(product_id,action_type,new_value)
VALUES(NEW.product_id,'New Product',NEW.name);
END$$
```

DELIMITER ;

-- 6 Prevent Deleting Active Products

DELIMITER \$\$

```
CREATE TRIGGER trg_no_delete
BEFORE DELETE ON prod14
FOR EACH ROW
BEGIN
IF OLD.stock_quantity>0 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT='Cannot delete product with stock';
END IF;
END$$
```

DELIMITER ;

-- 7 Track Deleted Products

DELIMITER \$\$

```
CREATE TRIGGER trg_delete_log
AFTER DELETE ON prod14
```

```
FOR EACH ROW
BEGIN
INSERT INTO audit14(product_id,action_type,old_value)
VALUES(OLD.product_id,'Product Deleted',OLD.name);
END$$
```

```
DELIMITER ;
-- 8 Minimum Price Enforcement
```

```
DELIMITER $$
```

```
CREATE TRIGGER trg_min_price
BEFORE INSERT ON prod14
FOR EACH ROW
BEGIN
IF NEW.price <1 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT='Minimum price is 1';
END IF;
END$$
```

```
DELIMITER ;
-- 9 Record Stock Increase
```

```
DELIMITER $$
```

```
CREATE TRIGGER trg_stock_increase
AFTER UPDATE ON prod14
FOR EACH ROW
BEGIN
IF NEW.stock_quantity > OLD.stock_quantity THEN
INSERT INTO audit14(product_id,action_type,old_value,new_value)
VALUES(OLD.product_id,'Stock Added',OLD.stock_quantity,NEW.stock_quantity);
END IF;
END$$
```

DELIMITER ;

-- 10 Prevent Weekend Price Change

DELIMITER \$\$

CREATE TRIGGER trg_weekend_block

BEFORE UPDATE ON prod14

FOR EACH ROW

BEGIN

IF DAYOFWEEK(CURDATE()) IN (1,7) AND OLD.price<>NEW.price THEN

SIGNAL SQLSTATE '45000'

SET MESSAGE_TEXT='No weekend price update';

END IF;

END\$\$

DELIMITER ;